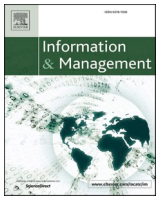
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Can B2B online communities reduce enterprises' default behavior?——From social network perspective

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ABSTRACT

Business-to-Business (B2B) online communities (OCs) facilitate enterprises to obtain additional resources through establishing interconnected networks. However, the existing literature lacks comprehensive insights into how the social network of enterprises affects their involuntary default. Based on social network theory, this study aims to investigate the influence of structural network embeddedness (i.e., density, structural holes, and centrality) on enterprises' involuntary default. The results show that these network characteristics have a direct impact on involuntary default, and these effects are moderated by role superposition.

1. Introduction

Involuntary default is considered an unintentional behavior in which an enterprise fails to repay its debts or fulfil its contractual obligations on time, usually due to external factors or uncontrollable circumstances [1]. In the past, researchers have identified factors related to involuntary default, such as economic difficulties, market changes, accidents and other unforeseeable objective environments [2]. In Customer-to-Customer online communities (C2C OCs), there are certain phenomena, such as the separation of information and objects, products and platforms, traders in physical space, and payment and commodity delivery [3,4], also contributing to consumers' default. In the Business-to-Business (B2B) online communities (OCs), the default of enterprises is usually involuntary, and the main reasons are as follows: First, unlike other OCs, their members are individuals, and their default is determined by individual motivations and capabilities. B2B OCs are comprised of businesspeople representing enterprises [5], and their default may be influenced by misunderstanding of contract terms, unexpected supply disruptions, or operational failures caused by market demand and industry changes [6–8]. For example, during the COVID-19 epidemic, the demand for medical supplies surged, and many enterprises fell into the predicament of insufficient production capacity. The failure to deliver on time led to an increase in the risk of default, which affects income and word of mouth of enterprises. Thus, they have high motivation to mitigate such default. Second, the platform will manage the default through formal and informal mechanisms [9]. These measures can damage the reputation of enterprises and threaten their long-term success, thus preventing enterprises from deliberately defaulting [10]. Therefore, enterprises in B2B OCs have very low motivation to intentionally default. Digitalization increases the dynamics and uncertainty of the environment, which requires enterprises to be more capable in meeting the increasingly complex market environment [11], thus reducing involuntary default.

An increasing number of enterprises actively participate in B2B OCs to obtain additional resources through establishing interconnected networks [12]. A social network embedded by enterprises affects the acquisition and utilization of resources and determines the richness of knowledge that enterprises can access [13,14]. This is also known as the influence mechanism of network embeddedness. The literature reveals that previous empirical research has focused on the effect of network embeddedness on behavior [15], the direct association of network embeddedness with positive outcomes of performance such as sales [16], knowledge contribution [17], and firm innovation [18]. Few empirical studies have focused on the relationship between network embeddedness and involuntary default in B2B OCs. Therefore, there is still a lack of systematic examination of how network embeddedness affects enterprises' involuntary default in B2B OCs.

Much literature on the social network has proposed that on the one hand, nodes with high network embeddedness have the advantage of using and controlling resources because they occupy a favorable position in the network [16] which enhances their ability and performance and influences the motivation and capabilities of enterprises behavior [19,20]. The Alibaba merchant community, an example of B2B OCs, provides opportunities for enterprises to build and extend their social

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networks. Enterprises in the network can obtain resource support, so as to quickly respond to market reactions and flexibly adjust delivery time and quantity. Therefore, the Alibaba merchant community can partially attribute its success to providing enterprises with the function of social interactions, thus promoting business transactions. The network embeddedness of enterprises within Alibaba is helpful to promote cooperation, information sharing, and resource acquisition among enterprises, thus reducing their involuntary default. On the other hand, owing to the limitation of cognitive resources [16], enterprises may not be able to screen out valuable professional knowledge from a large amount of diverse information, which may cause information overload [21]. Previous studies have confirmed that information overload is positively related to enterprise decision-making mistakes and potential risks [22,23]. Therefore, this paper proposes the bright and dark sides of network embeddedness on enterprises' involuntary default, hoping to provide new insights for enterprises to effectively enhance their capabilities and competitiveness in B2B OCs.

Furthermore, the dual commercial and social attributes of B2B OCs mean that enterprises not only play the role of businesspeople but also that of friends, which sets them apart from other OCs. Through information exchange and interactions leading to the development of relationships, enterprises may evolve into business partnerships or even friendships. The focal enterprise plays dual roles of businesspeople and friends in the specific seller-buyer relations, which is called role superposition. However, few studies consider the dual roles in B2B OCs, and more importantly, most research related to role theory shows that dual roles contribute to individuals' cognitive limitations and failure to meet the role expectations of group members [24], which leads to role conflict and negative effects [25]. In contrast, in the context of B2B OCs, role superposition may enable the focal enterprise to facilitate in-depth information exchange from a specific enterprise, which helps the focal enterprise obtain detailed information on supply and demand dynamics. When such detailed information from the specific dyadic relationship is combined with broad information from the network, they will complementarily influence the enterprise's ability to satisfy demand, thereby reducing default in the whole network and even those outside of the network. Therefore, this paper focuses on the potential positive spillover effect of role superposition, demonstrating that the mutual promotion between the two roles can have beneficial outcomes.

As stated in the introduction, this study focuses on a common social network in OCs, where multiple members reply to a post of a focal enterprise, aiming to examine the double-edged effect of network embeddedness on enterprises' involuntary default and the moderating effect of role superposition. We used the social network analysis (SNA) method to demonstrate and identify network characteristics, and the fixed-effects model to test the research hypotheses. Various robustness checks were conducted to verify the validity of our research results in the presence of potential collinearity and endogeneity. This study makes several contributions to the literature. First, this study contributes to the literature on default and B2B OCs by investigating the influence of structural network embeddedness on enterprises' involuntary default from network perspective. This paper also enriches theoretical research on the network embeddedness. Structural network embeddedness captures network structure characteristics (density, structural holes, and centrality) and their impact on network member performance [26]. However, few empirical studies have explored the double-edged effect of structural network embeddedness in reducing involuntary default. This study adopts an empirical approach to unravel the black box of whether structural network embeddedness is both beneficial and negative for reducing enterprises' involuntary default. Second, this paper extends the literature on role theory by proposing the potential spillover effect of role superposition. Previous research on role theory primarily focused on the negative impact of role conflict and role stress caused by dual roles [27,28] and rarely investigated its positive role in reducing enterprises' involuntary default. Our findings show that the effect of structural network embeddedness on enterprises' default behavior depends not only on network characteristics but also on role superposition. Moreover, dual roles are complementary in reducing default. Third, our study established enterprises' network based on post-reply links, rather than a self-reporting network, thus avoiding the problem of self-deviation.

2. Literature review and theoretical background

2.1. Business-to-business online communities (B2B OCs)

In recent years, OCs have been on the rise with the advancement of digital technology. B2B OCs are regarded as communities that facilitate both business transactions and social interactions for suppliers and buyers, where both buyers and sellers are enterprises [29]. B2B OCs offer more e-commerce functions than online social communities, such as Facebook, and incorporate greater social functions than other OCs related to e-commerce, such as Business-to-Consumer (B2C) OCs and Consumer-to-Consumer (C2C) OCs. As B2B OCs can promote sales through the exchange of information, knowledge, and experience, an increasing number of enterprises are actively participating in OCs [30,31]. However, information asymmetry in online virtual communities leads to misbehaviors, such as the generation and dissemination of misinformation in online health communities [32], the manipulation of personal information [33,34], and failure to fulfill promises in online crowdfunding communities [35].

Much research has contributed to the understanding of the drives of misbehaviors and how to manage them. In terms of motivations, Lin and Pursiainen [36] suggested that a lack of social capital may lead to misbehavior. Wang et al. [37] found that life dissatisfaction, perceived unfairness, inferiority feeling, and loneliness have a considerable effect on borrowers' default intentions. These studies conducted empirical tests through identifying social or psychological factors. However, B2B OCs encompass both social and commercial factors. B2B OCs not only facilitate business transactions directly but also indirectly increase sales by promoting social interactions. This dual commercial and social attribute means that enterprises not only play the role of businesspeople but also that of friends, which set them apart from other OCs. Regarding management, Zhao and Liu [33] investigated users' video-sharing social network and explored monitoring mechanisms to detect users' misbehavior. Liu et al. [38] suggested the friendships in individuals' social network can act as "pipes, prisms, and herding signals" in P2P lending. In the social network constructed by existing research, the nodes are individuals, and their behavior is motivated by individual interests. However, the nodes of social networks in B2B OCs are businesspeople representing enterprises that meet others in the same industry, interact on various issues, and conduct transactions to achieve business purposes

Table 1
The differences between B2B OCs, C2C OCs, and B2C OCs.

	C2C OCs	B2C OCs	**B2B OCs**
Author(s)	Thompson et al. (2016) Woo et al. (2024)	Pedeliento et al. (2020) Ozdemir et al. (2020)	Zhang et al. [5] Gharib et al. (2017)
Oriented	Information sharing	Enterprise's benefits	Business transactions
Example	Mafengwo; Airbnb	Dell community	Alibaba merchant community
Membership Member's ties	**individual** Consumer-consumer	**individual** Consumer-brand/ consumer	**enterprise** Buyer-seller
Function	Online interactions and transactions	Online interactions	Online interactions and transactions
Member's Role	**Single role: consumer**		**Dual roles: businesspeople & friends**

[39]. Their default is not only affected by their motivations and capability, but also by external environment such as market changes. Table 1 illustrates the differences between B2B OCs, C2C OCs, and B2C OCs. Given the unique characteristics of B2B OCs, it would be inappropriate to apply insights from other OCs to B2B OCs. Therefore, we should be clear about the enterprises' default in B2B OCs and how to restrain it, so as to provide implications for the transactions and interactions between enterprises.

2.2. Involuntary default

Involuntary default is considered an unintentional default behavior in which an enterprise fails to repay its debts or fulfil its contractual obligations on time, usually due to external factors or uncontrollable circumstances [1]. Involuntary default is not because the enterprise strategically chooses not to perform its obligations, but because of economic difficulties, market changes, accidents and other unforeseeable objective environments, it cannot fulfil its obligations as agreed to [2]. It is an operational management issue. The existing research has pointed out that intentional default is an immoral behavior, and debtor's involuntary default on loans because of the "can't pay" caused by sudden unemployment shock [40]. Economic uncertainty leads to a more complicated business environment, such as when an enterprise defaults on bonds, investors may lose confidence in the credit quality of the area, so the cost of local debt financing increases [41]. Financing difficulties worsen the liquidity of enterprises, cause operational difficulties, and eventually lead to involuntary default. Involuntary default within B2B OCs mean that an enterprise's unintentional failure to fulfil its obligations under the contract, which may involve failure to deliver goods according to the agreed-upon timelines and the products after the transaction are not in line with expectations [19,42].

The existing research has explored the factors that affect the enterprise's involuntary default. Financial difficulty is the most direct factor related to enterprises' default as financial resources, income, assets, and ownership are widely used to evaluate the probability of default [43]. Enterprises with good financial statements have the ability to cope with the risks brought by business failure, and the probability and risk of being forced to default are low [44,45]. In the era of big data, information and technology are the key aspects to improve the financial performance of enterprises. Acquiring and sharing information enables enterprises to quickly identify the trend of consumers' tastes and gain flexibility in responding to market changes [46]. Technical knowledge and skills can help enterprises improve their innovation output [47]. There are also studies showing that the macroeconomic environment can cause enterprises to fall into operational difficulties, which leads to involuntary default due to their inability to fulfil a contract. Chen et al. [48] pointed out that financial crisis, as an exogenous shock, has an adverse impact on enterprises' involuntary default. Stepankova and Teply [49] illustrated the massive impact of the COVID-19 crisis on probability of default, which is reflected in the real economy through the suspension of production, the closure of shops, and the reduction of income and expenditure. Keshavadasu [50] also explored several challenges faced by project development and operation, including regulatory and policy risks. Other literature discusses the importance of information, technology, and business knowledge to enterprise performance and competitive advantage, and think that the lack of these resources may lead to involuntary default [51,52]. Accordingly, it is very important for enterprises to improve their access to resources, manage the increasing amount of information and constraints posed by the external environment, and thus have a rich understanding of suppliers, needs, and environment [53].

Although some enterprises, due to abundant resources, are more likely to achieve innovative results, thus improving enterprise performance [54], other enterprises do not have rich resources, and are faced with the shortage of internal capabilities, which contributes to their involuntary default. Along with the rise of online communities, B2B OCs, a platform that can aggregate and integrate resources and realize resource allocation, provides possibilities for enterprises to obtain resource support [55]. In the B2B OCs, enterprises can establish social networks through social activities in order to obtain business information and expand social relations. Previous studies have shown that network structure can provide resource support and then affect performance such as sales [16], knowledge [17], and firm innovation [18]. Yet the impact of network structure on enterprises' involuntary default warrants further exploration.

2.3. Network embeddedness

Network embeddedness as proposed by Granovetter [20], refers to the organizational form of enterprise economic activities within a certain network relation. Network structural embeddedness and network relational embeddedness are fundamental concepts in the framework of network embeddedness [56]. B2B OCs allow members to use knowledge, exchange information, and share business experience through social networks, thereby accumulating resources and promoting the improvement of enterprise capabilities and performance. Therefore, these communities can be analyzed using social network theory, which focuses on the fact that a social network embedded by enterprises affects the acquisition and utilization of resources and determines the richness of knowledge that enterprise can access [13,14]. This is also known as the influence mechanism of network embeddedness. In the context of B2B OCs, enterprises have high motivation to mitigate default, and they may turn to social networks for support. The social network may provide resource support, such as supply dynamics information, market demand trends, and operational knowledge, to mitigate involuntary default. However, the role of networks in reducing involuntary default is not well understood within the context of B2B OCs. Existing literature on network embeddedness in OCs mostly focuses on social-oriented online communities with obvious strong relationship characteristics [57]. Therefore, relational embeddedness is suitable for exploring peer effects. However, B2B OCs do not exhibit strong relational networks but rather strong network structural characteristics, such as a large number of network nodes and a large network scale. So, we propose that network structural embeddedness can impact enterprises' involuntary default in B2B OCs.

Structural network embeddedness is defined as the impersonal configuration of linkages between network actors [58], focusing on network structure characteristics and their effects on network member performance. For example, nodes with high network embeddedness have the advantage of using and controlling resources because they occupy a favorable position in the network [16]. Previous literature on structural embeddedness included firm-level, pair-level, and network-level analyses [59–61]. In our context, we concentrate on the effect of firm-level embeddedness on enterprises' involuntary default. The reason is that, first, we focus on a focal enterprise's relationships with others in the network and how their network characteristics influence involuntary default. This concern emphasizes firm-level embeddedness, such as the focal enterprise's position in the network. Second, density, structural holes, and centrality are extensively used indicators of firm-level structural embeddedness [62–64]. Density captures the interconnections among enterprises in the network, structural holes represent information gaps, and centrality denotes the extent to which focal enterprises occupy a central position. These indicators are significantly for enterprises, directly affecting the extent of information and resources an enterprise can obtain from its network relations, and consequently, impacting performance. To fully understand the influence of network characteristics on enterprises' involuntary default, we systematically investigate the influence of network structural embeddedness on enterprises' involuntary default and describe how these three indicators affect the motivation and ability of enterprises to engage in default behavior.

3. Research model and hypotheses

3.1. Structural network embeddedness and enterprises' involuntary default

Structural network embeddedness pays more attention to the allocation of exchangeable resources in network structure and how to affect economic activities. Embeddedness implies that heterogeneous resources, specific knowledge, and information are embedded in the network [65]. The higher the structural embeddedness, the easier it will be for enterprises to obtain resources support and have stronger information acquisition and knowledge-sharing capabilities [15]. Previous studies have explored the influence of structural network embeddedness on enterprises performance. For instance, Mazzola et al. [58] argued that enterprises located in the center of the network can accumulate more knowledge and information, so they can turn this knowledge into further product innovation, while the presence of structural holes does not affect innovation performance. Su et al. [26] indicate that there is a positive relationship between network centrality and enterprise performance. The closer an enterprise is to the network center, the more it can seek new development opportunities for the enterprise and obtain the resources it urgently needs, so as to create new value and achieve the ultimate economic goal. Empirical evidence from Chen and Yang [13] indicates that network centrality and density positively affect sales through increasing consumer purchase intention. This means that enterprises occupy a very important position in the network, which directly affects their access to and use of network resources [15]. Most research is devoted to exploring the real networks of enterprises, such as trading networks and innovation networks, but ignores the influence of virtual social networks in B2B OCs.

According to previous literature, structural embeddedness of an enterprise is assessed using three key aspects of its position in the network. The first is density – the overall pattern of interconnections between enterprises – indicating the efficiency of information exchange [66]. The second is the presence of structural holes – the connection pattern among other enterprises in the focal enterprise's immediate neighborhood – highlighting the opportunity to act as bridges between other enterprises in the network [16]. The third is centrality – the direct connections of an enterprise in the network – reflecting the degree of resource access and information sharing [67]. These three characteristics are important aspects of the structural embeddedness network [68] and are highly related to the context of B2B OCs. Next, we will discuss how density, structural holes, and centrality affect enterprises' involuntary default. Fig. 1 illustrates the proposed research model based on the aforementioned theoretical background. To understand the relationships between several variables, we propose and examine six hypotheses.

3.1.1. Density

Network density is defined as the average strength of connections among units in a network, which reflects the closeness of connections between the nodes [64]. Enterprises maintain and continuously expand their online contacts with potential partners and establish friendships through social behavior. In such an enterprise-centered social network, the closeness of the relationship between the enterprise and other enterprises is called the network density of the focal enterprise [69]. An increase in density signifies a rise in interconnections between enterprises within the network [66].

Density may mitigate an enterprise's involuntary default. First, a dense network with high levels of interconnection helps to increase understanding of customer demand and flexibly responds to market demand, thus mitigating default. For instance, in a higher density network, the focal enterprise can easily contact the demand side, and can provide customized products and services according to the customer's needs, triggering or strengthening the completion of the transaction as expected [70]; or the focal enterprise can choose suppliers who provide the best materials and capabilities to reduce the risk of default caused by supply interruption [71], or can contact potential customers to understand their business needs and be able to meet potential market demand and reduce the possibility of default [72]. Second, compared with a loose network, a higher density network contributes to the rapid flow of information [73], allowing the focal enterprise to accumulate information and knowledge faster through dense interconnection [74, 75]. Enterprises that obtain fresh business information can make more informed decisions and respond faster to changes in their industry. Information advantage can bring development opportunities for enterprises, promote cooperation between enterprises and solve problems [76]. Through information exchange and cooperation in a dense network, the distortion of business information in the transmission process is avoided [16], so that enterprises can obtain new information quickly and effectively, and strengthen their ability to cope with the uncertainty of industry development. Consequently, in the context of social network, this paper proposes that high network density can reduce enterprises' involuntary default by interconnectedness, effectively understanding market demand, and obtaining business information in a timely fashion. Based on the above, we propose the following hypothesis:

H1. The higher density, the less likely enterprises to default.

3.1.2. Structural holes

Structural holes consider the unconnected situations presented by other enterprises that are directly connected to the focal enterprises in the network [63]. There are two advantages of resource support brought by structural holes. One is that structural holes capture the role of the focal enterprise as a broker [77], which means that the focal enterprise gains access to diverse information from the mutually unconnected enterprises in the network [78], and the other is that the focal enterprise occupying the structural hole plays the role of gatekeeper, i.e., has the ability to control the transmission and diffusion of information flow, and can promote or limit the interaction of information flow among other enterprises in the network [79]. Existing research has explored the positive influence of structural holes on enterprise performance, such as

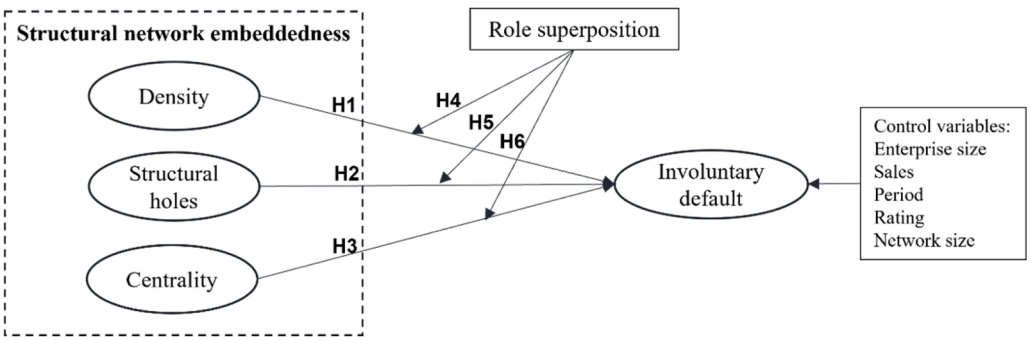


Fig. 1. Research model.

with innovation and technology improvement [77,80]. In particular, structural holes are effective for obtaining professional skills and knowledge and increasing profits. However, in B2B OCs, most enterprises are short on resources, scale, and operating experience. They are not able to screen out valuable professional knowledge from a large variety of information. Mallapragada et al. [16] suggest that structural holes may not be beneficial because they might create cognitive resource constraints on the enterprises. This leads to the dark side of structural holes, that is, information overload.

Information overload brought by structural holes may increase enterprises' involuntary default. First, information overload means enterprises need more time and energy to process and understand it [21]. With the increase of structural holes, the amount of information that enterprises need to deal with may exceed the information processing capacity [81]. Too much information will also distract managers, which will make them unable to concentrate on market demand and industry changes, and the production efficiency of enterprises may be affected, thus increasing the possibility of default. Furthermore, the information brought by structural holes is sometimes redundant [82], so it's easy for enterprise managers to confuse important information with irrelevant information, and it is impossible to accurately judge which information is valuable and useful [22]. This may lead to deviation in making important decisions and increase the risk of default. Finally, information overload means that some information is inaccurate or incomplete [83]. Due to the lack of effective resources, enterprises can't integrate and analyze all kinds of information, which may mean bad decisions are being made and can lead to potential risk factors [23], including for involuntary default. Therefore, we propose the following hypothesis:

H2. The more structural holes, the more likely enterprises are to default.

3.1.3. Centrality

Network centrality refers to the extent to which a node is connected to others and locates a central position in the social network [84]. Chen et al. [85] used network centrality to evaluate the influence of network effect in the P2P lending market and highlighted the importance of lender's network location. This paper discusses the influence of centrality on enterprises' involuntary default in three aspects. First, the higher the centrality, the more links an enterprise has with other enterprises, providing more opportunities for contact, communication, and obtaining resources, and the stronger its agglomeration and association ability with business elements [86], which affects the operational ability of enterprises and thus reduces the involuntary default. For instance, enterprises in the central position have access to a large amount of information, including successful and failed business experiences, which promotes enterprises to better understand market demand and industry changes [87]. By screening for the most suitable information, they may make more accurate choices and strategies. In addition, enterprises located in the central position are facing more trading opportunities [88]. Cooperation and trading with more enterprises can help the focal enterprise to constantly sum up business experience, promote the innovation of products and services [89], and enhance its sustainable operation ability. Second, enterprises with high centrality also have a greater influence within the network [90]. The focal enterprise with greater network influence is the maker of network rules, which determines that it has stronger control over network resources [91]. Through resource integration, economies of scale or synergistic effects can be realized, thus improving enterprise performance. Of course, network centrality also increases the possibility of getting help when enterprises are in trouble, thus reducing the risk of involuntary default. This view is common in loan decision-making based on social network [92]. Third, enterprises can gain the benefits arising from multiple knowledge channels and information sources, such as establishing competitive advantage, thus improving their performance. On the one hand, Cadden et al. [93] point out that existing knowledge sets can create stronger value-adding capabilities that allow an enterprise to obtain an advantage over its competitors. On the other hand, central position makes it easier for the focal enterprise to get in touch with new knowledge, and it is difficult for them to be imitated by their closest competitors, so that enterprise can gain a competitive advantage and improve enterprises' performance [94,95]. In general, in the context of the social network, this paper proposes that network centrality can reduce enterprises' involuntary default by improving operational ability, establishing influence, and cultivating competitive advantage. Therefore, we propose the following hypothesis:

H3. The higher centrality, the less likely enterprises to default.

3.2. Interaction between structural network embeddedness and role superposition

Role superposition refers to the focal enterprise playing dual roles of businesspeople and friends in the specific seller-buyer relations. The dual role theory holds that individuals and organizations can play the roles of businesspeople and friends at the same time in business relations [96]. In our context, the business transaction and social function of B2B OCs not only allows enterprises to conduct business transactions directly, but also allows enterprises to complete business transactions indirectly through social interaction, which is similar to relationship marketing [97]. Through common interests and values, enterprises can realize business cooperation and establish friendly relations in B2B OCs, thus strengthening mutual information and resource support. Specifically, when enterprises join B2B OCs, they initially act as businesspeople, and the purpose of interacting with other enterprises is to obtain more benefits. Structural network embeddedness endows enterprises with the ability to access information and resources from social networks. The focal enterprise establishes friendship through social interaction and business fairs with specific enterprises in the network; that is to say, the focal enterprise assumes the role of friends on the basis of the role of businesspeople. Business ties are often intertwined with friends, and the roles of friends and businesspeople coexist. According to the "reinforcement hypothesis" of role theory, individuals who perform responsibilities in multiple roles can get more resources to better realize development in many fields [98]. Recent research shows that the roles of employees in work and family are mutually reinforcing and positively affect their work and family quality [99,100]. Role superposition provides important market resources for B2B enterprises [101]. These resources are particularly useful, especially when the market environment changes dynamically. Role superposition enables the focal enterprise to communicate with specific enterprises in depth, which is helpful for the focal enterprise to obtain more detailed and accurate dynamic information of supply and demand [102]. Such information from a specific binary relationship combined with broad information from the network will complementarily influence the overall and comprehensive capabilities of focal enterprises, such as meeting market demand trends and responding to supply changes, thereby affecting the involuntary default in the whole network and even outside the network, that is, the spillover effect of role superposition.

3.2.1. Interaction of density, involuntary default, and role superposition

The higher the density, the greater overlap and interconnections among enterprises in the focal enterprise's network neighborhood [63], which means high communication efficiency between enterprises and it promotes information and the exchange of resources [13]. A dense network helps with better understanding of the changes in market demand and provides customers with satisfactory products and services. As enterprises change from the role of businessmen to the dual roles of businessmen and friends, role superposition helps to obtain resources from specific partners. Role superposition carries out detailed information sharing and in-depth business communication [103], which enhances enterprises' understanding of broad information obtained from the networks and further overcomes the adverse effects of resource constraints. We argue that role superposition strengthens the negative

effect of density on enterprises' involuntary default.

From the perspective of demand, when a retail enterprise assumes a dual role, it can understand the market demand and trend through in-depth communication with specific customers. At the same time, high network density means the focal enterprise has closer ties with other retail enterprises and suppliers in the network. If the information exchange between the focal enterprise and customers is sufficient and accurate, and the connection with other retail enterprises and suppliers is also very close, then the focal enterprise can adjust commodity inventory, price strategy, and sales channels more flexibly in the whole network and even outside the network [64,104]; for example, when the focal enterprise and customers are both business partners and friends and realize that the demand for a certain product will increase rapidly. In the case of high network density, contact with other retail enterprises and suppliers is very close, so that the focus enterprises can adjust order quantity and the purchase cycle to suppliers more quickly and are more prepared to meet market demand both inside and outside the network [105,106], thus reducing the potential demand-side default risk.

Additionally, previous studies have confirmed that dual relationships can help enterprises obtain important production information, such as raw materials, ingredients, and new methods of manufacturing [102]. On this basis, it is pointed out that the business and friend ties between the focal enterprise with specific enterprises is conducive to obtaining crucial market intelligence and selecting suppliers with excellent and efficient capabilities. Such in-depth information is very important to reduce the risk of involuntary default caused by sudden supply interruption, especially when the network is highly dense, enterprises overlap more, and information is broad. Therefore, we expect that the synergistic effect between density and dual roles can not only reduce the involuntary default of enterprises in the network, but also reduce the involuntary default of enterprises outside the network by improving the response ability of enterprises to changes in supply and demand. We propose the following hypothesis:

H4. The influence of density on enterprises' involuntary default is strengthened by role superposition.

3.2.2. Interaction of structural holes, involuntary default, and role superposition

The absence of structural holes means that the network is closed, where enterprises are interconnected and communication is smooth [16]. On the contrary, the existence of structural holes means that other enterprises in the network are connected to the focal enterprise but are not connected with each other [80]. Although structural holes can bring a variety of information, they will also cause communication barriers between enterprises, constrain cognitive resources, and cause information overload [16]. As we argued earlier, role superposition brings in-depth, detailed, and specific information, which can effectively guide the business strategy of the enterprise, including responding to changes in supply and demand.

With the focal enterprise changing from a single businesspeople role to a dual role of businesspeople and friend, communication is usually more open and informal and the information exchange between them is smoother [96]. Combined with increasing structural holes, role superposition makes it easier for enterprises to obtain key information and filter out irrelevant or excessive information. In this way, enterprises can focus more on the information related to their strategic objectives, support their decision-making inside and outside the network with a clearer perspective, alleviate the decision-making mistakes caused by information overload, and reduce the risk of involuntary default in the network or even outside the network. In addition, role superposition can promote the sharing of resources between the focal enterprise with specific enterprises. When two enterprises are not only business partners, but also friends, it is easier for them to support each other in terms of resources [107], such as sharing supply chain resources or providing temporary financial support. With the increase of structural holes, information overload makes the focal enterprise more dependent on resource sharing among friends. This flow of complementary resources helps the focal enterprise to maintain stability of operations in the face of emergencies or external pressures inside and outside network, thus reducing the risk of involuntary default caused by supply interruption or demand change [108]. Therefore, we expect that the synergistic effect between structural holes and role superposition can not only reduce the involuntary default of enterprises in the network, but also reduce the involuntary default of enterprises outside the network by complementary and targeted resource support. Based on this, we propose the following hypothesis:

H5. The effect of structural holes on enterprises' involuntary default is weakened by role superposition.

3.2.3. Interaction of centrality, involuntary default, and role superposition

Enterprises located in central positions have the ability to access more information, which can reduce their search and transaction costs [109]. In addition, the focal enterprise located in the center of the network can access more market trends, customer needs, and this broader information and resources have the potential for collaboration with network members and better performance in the network [5,89]. As mentioned above, high centrality leads to less involuntary default. We further posit that the central enterprise that can obtain in-depth, detailed, and specific information will gain additional benefits to strengthen the influence of centrality on the involuntary default of enterprises. The degree of obtaining this information is called the effect of role superposition, and it is considered as the ability of the focal enterprise to understand, absorb, and use information brought by dual roles, and leverage this information to its benefits [102,106]. Moreover, role superposition strengthens enterprises' understanding of market demand and supply changes by combining broad information, thus improving the ability and performance of enterprises, because the dual roles from a specific binary relationship will bring more specific, detailed, and in-depth information.

The combination of both broad information in the network and detailed information from role superposition are important for the focal enterprise to improve its capabilities [110]. Therefore, we argue that role superposition strengthens the negative effect of centrality on enterprises' involuntary default. For example, in the face of the sudden rise in raw material prices, the focal enterprise can find other sources of supply through the advantage of network location, and friendship can bring extra informal information to enterprises [38], help enterprises identify reliable suppliers and make better decisions [111]. It can also bring resource support, make suppliers more willing to provide discounts or flexible payment terms, thus improving the ability of the focal enterprise to cope with market demand and reducing involuntary default. Jraisat et al. [112] suggested that "relationship" is a key factor affecting enterprises performance, because it is related to information and resource sharing. In the presence of role superposition, a focus enterprise can not only benefit from the information advantage brought by high network centrality, but also obtain clear and specific information from the specific binary relationship, so as to improve its ability to cope with changes in market demand and supply, which may spill over to the outside of the network, thus reducing the involuntary default of enterprises within and outside the whole network. Based on this, we propose the following hypothesis:

H6. The influence of centrality on enterprises' involuntary default is strengthened by role superposition.

4. Methodology

4.1. Sample and data collection

To reduce the influence of self-reporting bias on model estimation, we use objective data from the Alibaba platform to test our model. The Alibaba merchant community is a third-party B2B online community that centers on transactions and relationship, providing a platform for

enterprises to exchange, share, and deliver information. The most obvious feature of the Alibaba merchant community is its integration of business and social functions, enabling enterprises to form a closed-loop trading experience in B2B OCs and promoting friendship among enterprises based on social interactions. Focused topics, lots of posts, rich data, and interactions among community members are the reasons why we chose it. Similar to Twitter, the Alibaba merchant community encourages enterprises to establish social relations through information exchange and sharing [5], with a primary goal of facilitating business transactions. There are many posts in the community, citing experiences with topics from the mainstream to the obscure. Enterprises can publish content by posting media, and other enterprises can reply to the posts, establishing a relationship between the enterprises. The enterprises' profile displays both their social and business characteristics (Fig. 2).

We collected the data in three phases. In phase one, we collected the "post-reply" relationships among enterprises to capture the structural network embeddedness of focal enterprises. We choose reliable posts and comments to reduce the deviation caused by invalid information (as shown in Fig. 3). Finally, 542 posters and 2025 replies were ~~crawled~~ (excluding multiple responses from one enterprise to another to reduce sample repeatability and accuracy of our results). Based on this, we established an incidence matrix where the rows represent the posting enterprises, and the columns represent the replying enterprises. In our research, this is a 542*2025 matrix. If enterprise *i* comments on the post of enterprise *j*, then there is a relationship between them, and we record it as 1, otherwise, we do not mark it (as shown in Fig. 4). We use a two-model social network analysis method to build a graph of the network where the network nodes represent the posts and replies by enterprises within the network, and the edge represents the "post-reply" relationship between the two nodes (as shown in Fig. 5). This enables us to calculate the structural network embeddedness of focal enterprises. One month after the first round of data collection, we crawled the control variables such as sales of focal enterprises from their profiles. A month later, we continued to collect enterprises' default information. Eventually, we matched the data across the three phases based on enterprise names and obtain all samples information. The purpose of collecting data in stages was to form a lag period between dependent and independent variables, thus avoiding endogeneity caused by the causality between independent and dependent variables [17].

4.2. The definition and measurement of variables

Dependent variable: Involuntary Default Behavior (*DB*). We focus on default trading behavior, mainly including transactions and product quality violations. Transaction violation refers some enterprises on Alibaba platform failing to deliver or pay on time as agreed due to unforeseen factors, which affects the rights and interests of buyers and/or sellers. Product quality violations refer to products provided by the enterprise that are inconsistent with the contents of the contract, which may be caused by the shortage of upstream raw materials resulting in the seller changing raw materials in order to trade on time. For these violations, the Alibaba platform gives enterprises a severe warning, restricts their rights, closes the store depending on the circumstances, and publishes enterprises' default behavior on their website. We use dummy variables to measure enterprises' involuntary default, using a value of 1 when the enterprise has defaulted and 0 otherwise. As mentioned before, we used lagged default data to alleviate endogeneity.

Independent Variables: Density (*DE*). The calculation formula for network density is the actual number of relationships in the focal enterprise's network divided by the theoretical maximum number of relationships (Liu et al., 2019). In an undirected network with *n* actors, if the actual number of relationships is *m*, then the equation for *DE* is:

$$DE_i = 2m/[n(n-1)]$$

The value range for the network density is between 0 and 1. If each enterprise in the network is connected to every other enterprise, then *DE* = 1 [16].

Structural Holes (*SH*). As per Mallapragada et al. [113] and Malm et al. [114], we used constraint to measure structural holes. In the following equation, p_{ij} represents the proportion of direct connection between node *i* and *j* in all connections of *I* in the network, and $p_{iq}p_{qj}$ represents the proportion of indirect connections between node *i* and *j* in all connections of *i* in the network:

$$SH_i = \left(p_{ij} + \sum_q p_{iq}p_{qj}\right)^2 \quad q \neq i,j.$$

Centrality (*CT*). Network scholars develop several centrality measures, such as degree, closeness, betweenness, eigenvectors, and flow betweenness, to measure network centrality [115]. We use degree centrality, which is the most-used measure for network centrality, to measure enterprises' positions in their social network. Degree centrality in a social network refers to the number of direct connections between a given node and other nodes [116]. In this paper, we use degree centrality to describe the number of other enterprises in the social network with which an enterprise has a social relationship (e.g., comments or replies). High-degree centrality means that the enterprise occupies a favored network position. In the following equation, *j* refers to an enterprise directly connected to enterprise *i*, and the formulation of degree centrality for enterprise *i* is as follows:

$$D_i = \sum_{j=1}^{n} j$$

Moderator: Role superposition (*RS*). In our context, the focal enterprise actively participates in and uses the market influence and resources of other enterprises in their transactions with other enterprises, which is considered the same as the businesspeople role of the focal enterprise. On this basis, it shows a deeper interactive relationship, which not only follows the strategic decision and market dynamics of enterprises, but also is followed by enterprises, which means that the focal enterprise assumes the role of friend in addition to the role of businessperson, that is, role superposition. Role superposition refers to an enterprise going from solely business to a dual role of businesspeople and friends. *RS* measured by judging the relationships of transaction and friendship between enterprises. We match the list of enterprises' followers, followees, and buyers. If an enterprise has a friendship relationship with the focal enterprise but has no business transaction records, or only has business transactions but no friendship relationship, *RS* is recorded as 0. Otherwise, *RS* is recorded as 1.

Control Variables: This paper explores the influence of structural network embeddedness characteristics on enterprises' involuntary default, which is soft information that affects enterprises' credit [117]. Therefore, the hard information that affects enterprises' credit should be controlled. Based on the literature [118,119], this paper controls several variables. (1) Enterprises' size (*EntSZ*). In this study, sample enterprises are divided into two categories according to the number of employees. With the number of employees less than 51, size = 0; with the number of employees greater than or equal to 51, size = 1. (2) Sales (*SA*). Sales significantly affect the cash flow and financial ability of an enterprise [120], which is the direct financial factor affecting the credit of an enterprise [121]. The total number of online sales completed by the enterprise is measured by the sales performance, which may affect the enterprise's default behavior. (3) Period (*PE*). The period of an enterprise indicates its experience in the Alibaba platform [122], which is highly-related to its development ability in the enterprise credit evaluation model. Therefore, the operating time may affect the enterprises' default behavior. In our context, period refers to the number of days from the enterprises' registration date in the Alibaba merchant community to the date when we obtain the data [123]. (4) Rating (*RT*). The Alibaba merchant community has a rating system, similar to the consumer evaluation system in C2C transactions, which is used to rate the

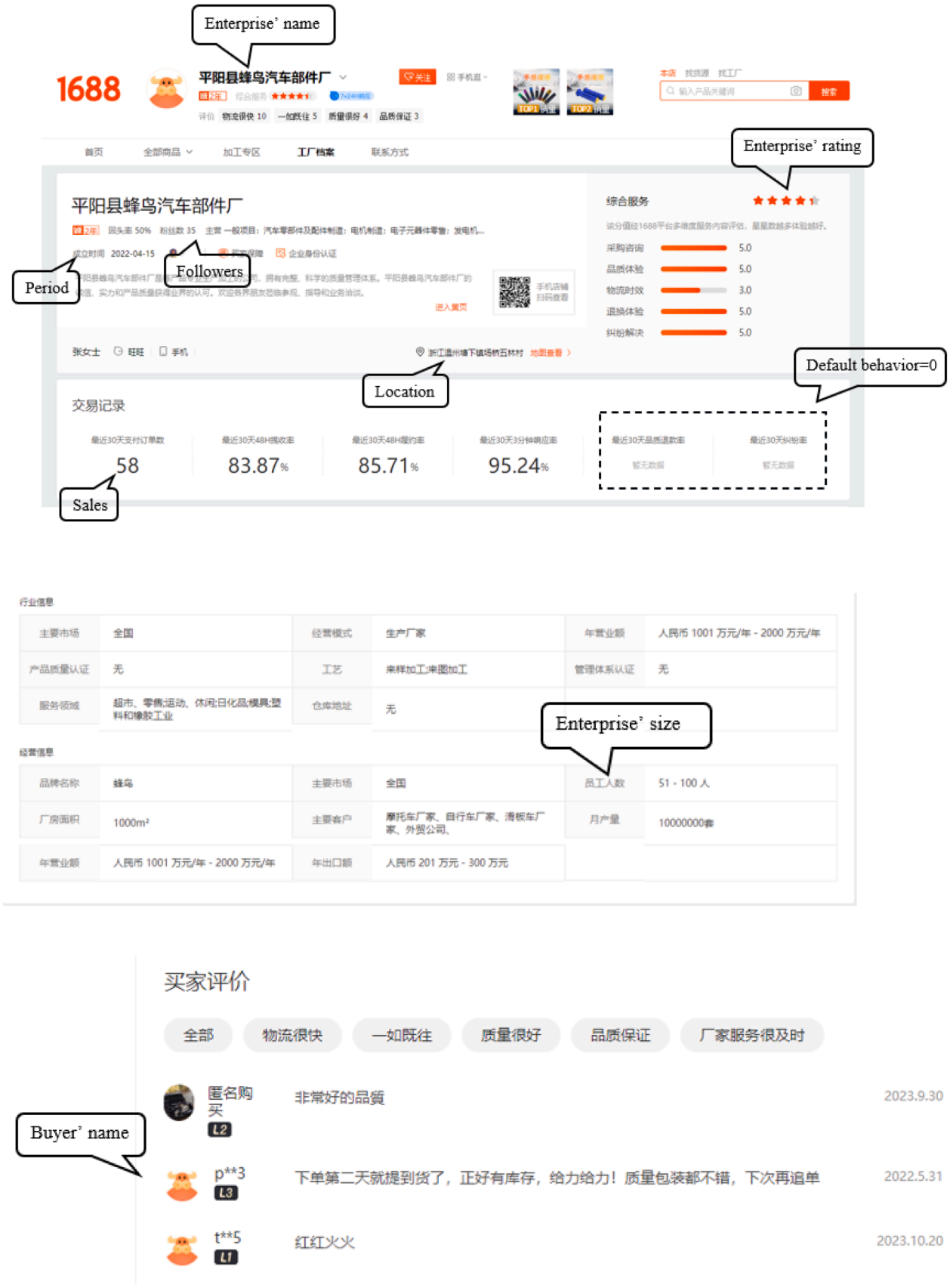


Fig. 2. Screenshot of an enterprise's profile.



Fig. 3. Screenshot of "excellent" label.

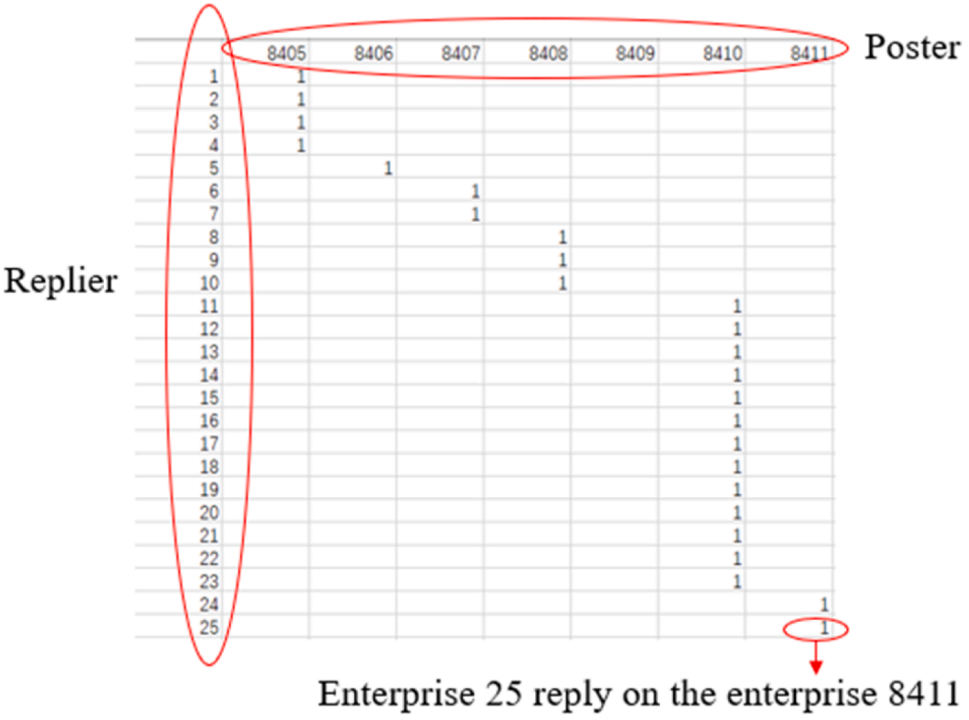


Fig. 4. The sample of an incidence matrix.

products and services of enterprises. The rating includes five grades (1 to 5), a higher rating indicates that the products provided by the enterprise are reliable in quality and professional in service, which makes customers satisfied to some extent. Previous literature has shown the negative relationship between rating and default haviour [124,125]. (5) Network size (*NetSZ*). Network size is a significant indicator affecting enterprises' performance [126]. We use the focal enterprise's network size using the total number of directed and undirected actors in its social network. (6) Industry (*IND*). We use 1–20 to represent 20 industries in the Alibaba merchant community.

4.3. Econometric models

To test hypotheses 1–3, we established the following econometric model:

$$DefaultBehavior_{it} = \alpha + \beta_1 Density_{i,t-1} + \beta_2 StructuralHoles_{i,t-1} + \beta_3 Centrality_{i,t-1} + \delta X_{it} + \mu_i + \nu_t + \varepsilon_{it} \quad (1)$$

In Formula 1, the subscripts i and t respectively represent an enterprise and a month. The independent variable is the enterprise's involuntary default behavior ($DefaultBehavior_{it}$), and the main dependent variables are network density ($Density_{i,t-1}$), structural holes ($StructuralHoles_{i,t-1}$), and centrality ($Centrality_{i,t-1}$) of the previous period. The reason why the lag value of dependent variables is used is to control endogenous correlation. We pay attention to the positive and negative coefficients β_1, β_2, and β_3. If β is negative and significant, it shows that the network structure has an inhibitory effect on the involuntary default of enterprises. X*it* is a vector composed of control variables such as enterprise size (*EntSZ*), δ is its coefficient vector, α and ε_{it} are constant and random disturbance terms respectively, μ_i and ν_t are enterprise characteristics and time-fixed effects respectively.

To test hypotheses 4–6, we established the following econometric model:

$$DefaultBehavior_{it} = \alpha + \gamma_1 Density_{i,t-1} + \gamma_2 StructualHoles_{i,t-1} + \gamma_3 Centrality_{i,t-1} + \gamma_4 RoleSuperposition_{i,t-1} + \gamma_5 Density_{i,t-1} * RoleSuperposition_{i,t-1} + \gamma_6 StructualHoles_{i,t-1} * RoleSuperposition_{i,t-1} + \gamma_7 Centrality_{i,t-1} * RoleSuperposition_{i,t-1} + \delta X_{it} + \mu_i + \nu_t + \varepsilon_{it} \quad (2)$$

In Formula 2, the independent variable is the enterprise's involuntary default ($DefaultBehavior_{it}$), and the main dependent variables are network density ($Density_{i,t-1}$), structural holes ($StructuralHoles_{i,t-1}$), centrality ($Centrality_{i,t-1}$), and role superposition ($RoleSuperpositiony_{i,t-1}$) as well as their interaction terms ($Density_{i,t-1}$*$RoleSuperpositiony_{i,t-1}$, $StructuralHoles_{i,t-1}$* $RoleSuperpositiony_{i,t-1}$, $Centrality_{i,t-1}$* $RoleSuperpositiony_{i,t-1}$). We use the lag value of role superposition in order to control endogenous correlation. We pay attention to the estimation coefficient γ_5, γ_6, and γ_7, which reflect whether the influence of network structure on enterprise involuntary default is more obvious in enterprises playing dual roles. X*it*, δ, α, ε_{it}, μ_i, and ν_t have the same meanings as in Formula 1.

We used a binary selection model for processing. In a linear probability model, the algorithm is very basic and the method is very extensible, so it can be estimated in many ways to ensure robustness and can fully cope with external shocks. We employed a high dimension fixed-effects model that controls for enterprises and time for processing 1 and 2, and deconcentrate independent variables and moderator variable to avoid multicollinearity.

4.4. Endogeneity issues

In this paper, homophily is the main source of endogenous problems. Homophily suggests that individuals are likely to form ties with people who are similar to them [127]. Under the influence of homophily, the behavior of social network members may be caused by their intrinsic similarities [128]. Based on previous studies about social networks, we adopt three methods to control homophily. First, control the observable factors. Godinho de Matos et al. [129] treated homophily as a factor to be controlled for. In the research of OCs, there are usually controls for member tenure and rank to alleviate the influence of homophily [130]. In our context, we regarded the days from the date when the enterprise is registered in the Alibaba merchant community as the tenure. The enterprise rating similar to reputation is taken as a control variable to control the influence of homogeneity [17]. In addition, rating, industry, and size as enterprises characteristics also are controlled. Second, we use longitudinal data. As per Noel and Nyhan [131], we collect data in three stages: the "post-reply" relationship and network structure data in the first period, the enterprise attributes and transaction data in the second period, and the default data in the third period. Such data can control the endogenous problems caused by homophily and alleviate reverse causality. Third, we employ a fixed effect model. Nair et al. [132] controlled the time fixed effect and physician fixed effect to eliminate the influence of homophily caused by physicians and doctors having similar "tastes" for prescription. Similarly, we control the enterprise and time fixed effects. These approaches could help to eliminate endogenous problems caused by homophily. In addition, creditable enterprises may have more "friends", so it is more likely to play dual roles. There may exist reverse causality issues in the model. We used generalized method of moments (GMM) to conduct regression analysis of equation 2. Specifically, when using the GMM, the lag period of the independent variable is added to the right of equation 2 to control the possible the of autocorrelation of the independent variable, and simultaneously control the current value and lag period of dependent variables.

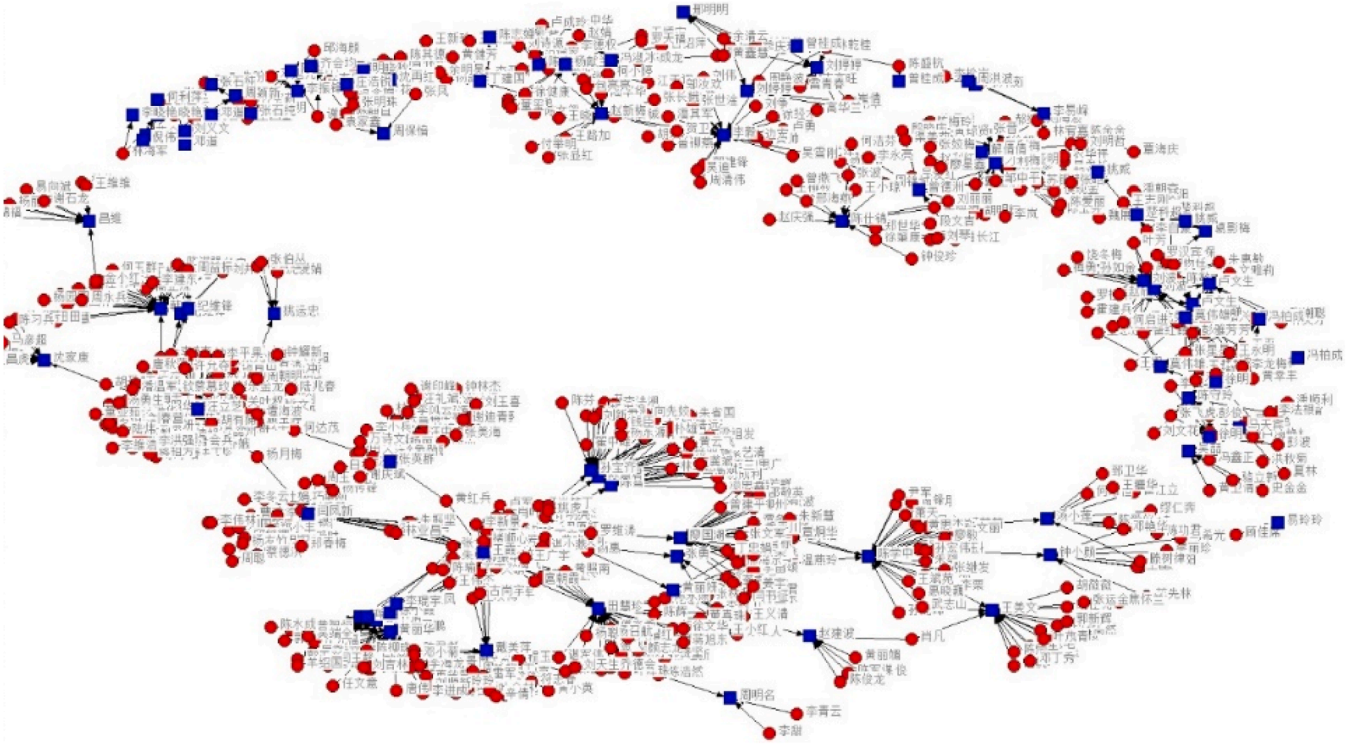


Fig. 5. An example: Social network graph.
Blue Squares: Enterprises that published posts; Red Circles: Enterprises that published comments; Line: Red circles (Enterprises) commented on blue circles (Enterprises).

Table 2
The distribution of sample.

	Distribution	Frequency	Ratio		Distribution	Frequency	Ratio
Size (the number of employees)	More than 0 and less than 50	328	60.5%	Industry	Costume	38	7.0%
	More than or equal to 51	214	39.5%		Mechanical equipment	27	5.0%
Age (years)	More than 0 and less than 3	53	9.8%		Packaging and printing	31	5.7%
	More than or equal to 3 and less than 5	343	63.3%		Medical maintenance	24	4.4%
	More than 5	146	26.9%		Communication digital	25	4.6%
Location	Southeast coast	275	50.7%		Food and beverage	28	5.2%
	North and Northeast	154	28.4%		Building materials	24	4.4%
	Middle	97	17.9%		hardware tool	33	6.0%
	The west	16	3.0%		Textile	29	5.3%
Industry	Electrician electricity	26	4.8%		Chemical	28	5.2%
	Safety protection	19	3.5%		Rubber and plastic	20	3.7%
	Home	28	5.2%		Automobile and motorcycle	30	5.5%
	Small commodities	14	2.6%		Agriculture	28	5.2%
	Flea market	34	6.3%		Energy and mineral resources	28	5.2%
	Other	28	5.2%				

Table 3
Descriptive Statistics.

Variables	Mean	SD	Min	Max	Median
DB	.493	.500	0	1	0
DE	.009	.007	.003	.051	.006
SH	.883	.865	.062	3.645	.819
CT	356.254	353.804	4	1436	231
SA	605.238	1089.35	0	9982	150.5
EntSZ	.395	.489	0	1	0
PE	1602.934	963.539	76	4895	1215
RT	3.474	.826	1	5	4
NetSZ	1527.449	762.241	700	3799	1279
IND	10.174	5.862	1	20	10
RS	.322	.467	0	1	0

5. Results

5.1. Descriptive data statistics

Table 2 shows the distribution of sample size, age, location, and industry. The descriptive statistics from the collected data (Table 3) show that the standard deviation (0.500) of the variable involuntary default of the sampled enterprises is greater than the average value (0.493), which indicates that there are differences in the enterprises' involuntary default. Because most variables are distributed to the right, we adopt the logarithmic form of variables in the subsequent analysis. To avoid multicollinearity, we test the correlations between variables before performing a regression test on the data in this paper. Table 4 shows that the absolute value of each correlation coefficient is < 0.8 and the maximum value of VIF is 1.123, which avoids collinearity in model estimation to some extent.

5.2. Regression results

Based on the above data, we use a linear probability model in our analyses. Column 1 in Table 5 shows the results with only control variables. Column 2 shows that social network density is negatively correlated with enterprises' involuntary default at a 1% level of significance, which indicates that the larger the network density, the less the enterprises involuntary default. The correlation coefficient between structural holes and enterprises' involuntary default is 0.062, which is significantly positive at a 1% level of significance. This result indicates that more structural holes provide limited resources for enterprises; therefore, enterprises are likely to be forced to default. Network centrality is negatively correlated with involuntary default at a 1% level of significance, and the coefficient is -0.109. Network centrality means that enterprises can directly obtain the available information and resources. Therefore, the more central an enterprise's network position is, the more resources it can use, and the more opportunities it has to enhance its operational capability and establish a competitive advantage, which effectively reduces their involuntary default. In general, our H1, H2, and H3 are supported.

5.3. Moderating effects

Based on our hypotheses, we consider that the dual roles enterprises play have a positive moderating effect on the three variables. To test the moderating effect, we introduce the following interaction terms: *DE *RS*, *SH *RS*, and *CT *RS*. In order to facilitate the statistical interpretation of independent variable coefficients and eliminate the influence of multiple collinearity problems, we centralize independent variables, moderator, and interaction terms. Column 3 in Table 5 shows the results. Role superposition shows a moderating role in the relationship between density or structural holes and enterprises' involuntary default (β_{DE*RS}= -0,052, t = -2.800, p < 0.01; β_{SH*RS}= -0.027, t = -1.940, p < 0.01). The coefficient between *CT *RS* and enterprises' involuntary default is

Table 4
Matrix of correlations and VIF.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	VIF
(1) *DB*	1.000											
(2) *DE*	-0.182	1.000										1.053
(3) *SH*	0.120	-0.037	1.000									1.135
(4) *CT*	-0.174	0.046	-0.106	1.000								1.034
(5) *EntSZ*	-0.103	0.087	0.051	-0.120	1.000							1.029
(6) *SA*	-0.001	-0.039	-0.060	0.021	-0.009	1.000						1.013
(7) *PE*	0.014	-0.141	0.297	-0.036	0.039	-0.046	1.000					1.123
(8) *RT*	-0.009	0.024	-0.010	-0.005	0.033	0.012	0.041	1.000				1.004
(9) *NetSZ*	-0.027	0.041	-0.030	0.050	0.023	0.007	-0.020	0.007	1.000			1.006
(10) *IND*	0.034	0.031	0.023	0.055	0.005	-0.037	0.081	0.093	0.006	1.000		1.030
(11) *RS*	-0.103	0.138	-0.152	0.070	0.027	-0.070	-0.062	0.012	0.028	0.099	1.000	1.052

Table 5
The results of regression analysis.

	Column (1)		Column (2)		Column (3)	
	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.
EntSZ	-0.259	0.205	-0.299**	0.128	-0.335***	0.139
SA	0.004	0.003	0.003	0.003	0.001	0.003
PE	-0.011	0.060	-0.104*	0.060	-0.090	0.058
RT	-0.046**	0.022	-0.042*	0.022	-0.043**	0.022
NetSZ	-0.019	0.013	-0.016	0.013	-0.299***	0.114
IND	-0.048	0.112	-0.074	0.109	-0.074	0.108
DE			-0.057***	0.009	-0.035***	0.011
SH			0.062***	0.016	0.068***	0.016
CT			-0.109***	0.012	-0.102***	0.013
RS					-0.303***	0.112
*DE*RS*					-0.052***	0.017
*SH*RS*					-0.027***	0.012
*CT*RS*					-0.005	0.012
Time FE	Yes		Yes		Yes	
Enterprise FE	Yes		Yes		Yes	
Adj R-squared	0.217		0.268		0.275	

*** p<0.01,
** p<0.05,
* p<0.1

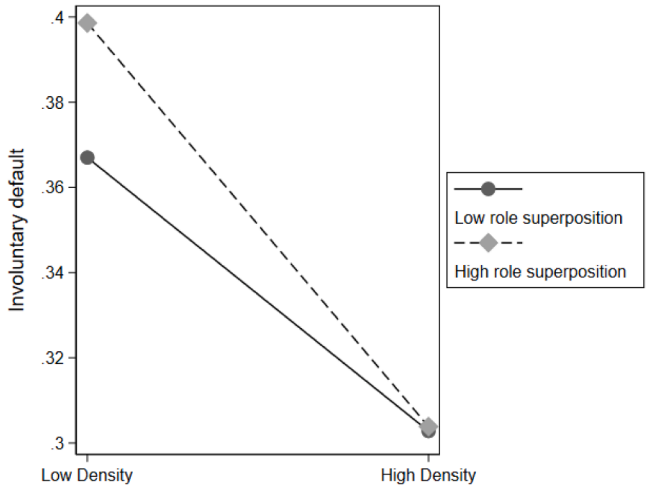


Fig. 6. The moderating effect of role superposition on the effect of density on involuntary default.

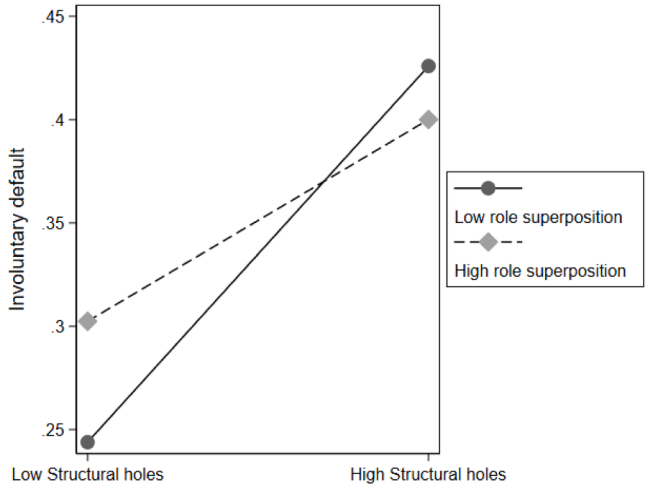


Fig. 7. The moderating effect of role superposition on the effect of structural holes on involuntary default.

Table 6
Robustness checks: Adding control variable.

	(1)		(2)		(3)	
	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.
EntSZ	-0.273	0.205	-0.302***	0.129	-0.312***	0.134
SA	0.003	0.003	0.003	0.003	0.001	0.003
PE	-0.011	0.060	-0.104***	0.060	-0.083	0.058
RT	-0.046**	0.022	-0.042***	0.022	-0.043**	0.022
NetSZ	-0.018	0.013	-0.016	0.123	-0.016	0.013
IND	-0.048	0.112	-0.074	0.109	-0.010	0.017
AT	0.016	0.012	0.004	0.012	-0.054***	0.017
DE			-0.057***	0.009	-0.034***	0.011
SH			0.062***	0.016	0.069***	0.016
CT			-0.109***	0.012	-0.104***	0.013
RS					-0.354***	0.115
*DE*RS*					-0.049***	0.018
*SH*RS*					-0.031**	0.014
*CT*RS*					-0.003	0.012
Time FE	Yes		Yes		Yes	
Enterprise FE	Yes		Yes		Yes	
Adj R-squared	0.217		0.268		0.277	

*** p<0.01,
** p<0.05,
* p<0.1

negative but non-significant (β_{CT*RS}= -0.005, t = -0.510, p > 0.1), which indicates the effect of role superposition is not valid. Thus, H4 and H5 are supported and H6 is not supported. Figs. 6 and 7 show the moderating effect of role superposition on the impact of density and structural holes on involuntary default.

5.4. Robustness checks

In an earlier analysis, we use the study enterprises' size, sales, network size, period and rating as the control variables. Furthermore, we add the social activities (*AT*) as a control variable to check the robustness of the results in this paper. *AT* refers to the number of social behaviors measured by the number of enterprises' posts and questions. Social behavior has been considered in the credit evaluation model [38, 133]. Chen and Li [134] show that the more social activities, the better corporate credit. Social activities facilitate the accumulation of social capital, which is regarded as collateral that affects enterprises' default behavior. Therefore, *AT* can be used as a control variable. Results are

Table 7
Robustness checks: Alternative method.

	(1)		(2)	
	Coef.	S.E.	Coef.	S.E.
EntSZ	-2.571	1.762	-2.583	1.766
SA	0.013	0.022	0.004	0.023
PE	-0.626	0.409	-0.360	0.425
RT	-0.339*	0.174	-0.366**	0.176
IntSZ	-0.143	0.102	-0.135	0.103
IND	-0.565	20.759	-0.616	36.090
AT	0.012	0.091	-0.421***	0.129
DE	-0.431***	0.069	-0.218**	0.086
SH	0.445***	0.098	0.502***	0.102
CT	-0.812***	0.088	-0.794***	0.095
RS			-3.454***	0.984
*DE*RS*			-0.500***	0.147
*SH*RS*			-0.283***	0.098
*CT*RS*			-0.005	0.105
Time FE	Yes		Yes	
Enterprise FE	Yes		Yes	
Pseudo r- squared	0.083		0.102	
Log lokelihood	-1000.983		-980.122	
LR chi- squared	181.753***		223.474***	

*** p<0.01,
** p<0.05,
* p<0.1

Table 8
Robustness checks: Alternative measurement.

	Coef.	S.E.
EntSZ	-0.312**	0.127
SA	0.002	0.003
PE	-0.119*	0.062
RT	-0.044**	0.022
NetSZ	-0.015	0.013
IND	-0.074	0.109
AT	0.009	0.014
DE	-0.053***	0.010
SH	0.066***	0.016
CT	-0.105***	0.012
RS	-0.004	0.004
*DE*RS*	-0.001*	0.001
*SH*RS*	-0.002***	0.001
*CT*RS*	-0.001	0.001
Time FE	Yes	
Enterprise FE	Yes	
Adj R-squared	0.270	

*** p<0.01,
** p<0.05,
* p<0.1

Table 9
Results of the GMM analysis.

	Diff-GMM		Sys-GMM	
	Coef.	S.E.	Coef.	S.E.
Lag_*DB*	-0.284***	0.020	-0.422***	0.017
EntSZ	0.734**	0.350	-0.010	0.279
SA	0.003	0.003	0.002	0.003
PE	0.015	0.073	0.033	0.076
RT	0.011	0.032	0.024	0.031
NetSZ	-0.009	0.013	-0.012	0.012
IND	-0.044	0.051	-0.049	0.118
DE	-0.032***	0.012	-0.029**	0.011
SH	0.070***	0.017	0.050***	0.016
CT	-0.143***	0.012	-0.129***	0.011
RS	-0.351**	0.140	-0.263**	0.130
*DE*RS*	-0.054***	0.020	-0.041**	0.018
*SH*RS*	-0.047***	0.017	-0.043***	0.016
*CT*RS*	-0.001	0.015	-0.002	0.014
Sargan	No overidentifying		No overidentifying	
Wald	687.725***		1274.725***	

*** p<0.01,
** p<0.05,
* p<0.1

shown in Table 6. For the control variables, column 1 shows that enterprises with higher third-party rating in B2B OCs were less likely to default. Column 2 is the main effect models and the column 3 represents the moderating effect models. The significance of the coefficients was not changed, indicating our model is robust.

Additionally, we conducted the logit regression analyses to test our model. In Table 7, column 1 is the main effect models, and the column 2 represents the moderating effect models. Most of the effects that we studied are quantitatively consistent with our main findings. The effects are quantitatively consistent with our main findings; therefore, we can be confident that the results of our analysis are solid and robust.

We also changed how to measure the moderator variable. We used the number of friendship relationships in enterprises to measure the degree of *RS*. The more friends, the greater the *RS*. Compared to enterprises with small *RS*, enterprises with large *RS* are more likely to form trust relationships. Table 8 shows that our results are robust.

We grouped the data according to whether the enterprise plays a dual role. When *RS*=0, the average value of *DB* is 0.528; *RS*=1, the average value of *DB* is 0.418, so we can preliminarily infer that the friendship within enterprises is not caused by *DB*, which reduces the possibility of reverse causality between them. In addition, we also used the generalized moment method (GMM) to alleviate reverse causality. Table 9 shows the results of equation 2 in differential GMM and systematic GMM regression. In GMM analysis, we add the lag period of the independent variable as a dependent variable to control the possibility of autocorrelation of the independent variable and control the current value and lag period of the independent variables at the same time.

6. Discussion and conclusion

6.1. Key findings

We explore the influence of structural network embeddedness on enterprises' involuntary default in B2B OCs. Our results show that structural network embeddedness plays an important role in helping to reduce enterprises' involuntary default. Network density is negatively correlated with enterprises' involuntary default. This finding is consistent with existing research on the bright sides of the density [63,66]. The initial role of an enterprise is to be businesspeople in the B2B OCs, and the dense network promotes the interaction between businesspeople [13], thus enhancing the trust between enterprises and reducing bad behavior [135]. As for structural holes, these have played a negative role. Literature on the effect of structural holes on the performance of members has presented strong evidence [58,114]. Although structural holes can bring a variety of information, they will also cause communication barriers between enterprises, constrain cognitive resources, and cause information overload [16]. The positive relationships between network centrality and behavior and performance of enterprises have reached a consensus [136,137].

Furthermore, our findings suggest that the dual roles of enterprises partially influence the relationship between structural network embeddedness and enterprises' involuntary default. Specifically, we observed that the effect of density on enterprises' involuntary default was strengthened by enterprises' dual roles, while the effect of structural holes was weakened by dual roles. These findings provide additional support for previous studies on the role theory. Enterprises initially operate as businesspeople and then establish social relations based on business ties, assuming the dual role of a businessperson and friend. Although the actions of businesspeople are typically driven by the principle of maximizing interests [96], the establishment of friendship may lead enterprises to provide valuable resources, even at the risk of short-term interests, in order to foster long-term relationships [138]. Therefore, the role of friend can promote the role of businesspeople, facilitating reciprocity, mutual benefit, and the sharing of industry information and experiences. Ultimately, this can lead to easier access to trading opportunities and improved enterprise performance. In general, the dual roles of enterprises can promote the exchange of specific binary relations, obtain more in-depth and detailed information, complement each other with broad resources in the network, and affect the capability of enterprises, thereby affecting the influence of network density and structural holes on enterprises' involuntary default.

Surprisingly, however, the dual roles of enterprises did not show a significant effect on the relationship between centrality and enterprises' involuntary default. This suggests the contingency condition for the moderation effect of enterprises' dual roles. Higher centrality means that enterprises occupy a favorable position, potentially due to other enterprises actively engaging in the focal enterprise's posts. This may be crucial for the role superposition of enterprises because more relationships contribute to centrality. However, enterprises with dual roles may not prioritize centrality. In the case of high network centrality, enterprises may have more resources and information advantages. This should reduce the possibility of involuntary default, but if the enterprise has a dual role in the network, the potential opposite effects of role superposition may result in an insignificant moderating effect. This can be explained by the influence of role expectations. As businesspeople, the purpose of an enterprise is to conduct business transactions, and the role of friend requires consideration of the interests of others perhaps

even over one's own [138], making it difficult for the enterprise to balance the expectations of two different roles [139]. Detailed information from dual roles cannot play a complementary role in the broad and abundant information brought by high network centrality, which is useless and even a side effect. Enterprises with role superposition may pay more attention to the higher status, because there may be little demand for them to pursue the dual relationship of businesspeople and friends, but they pay more attention to improving network centrality to improve operational capabilities, enhance influence and establish competitive advantages.

6.2. Theoretical contributions

Our study makes several theoretical contributions to the literature. First, this paper focuses on the enterprises' involuntary default in B2B OCs and puts forward a theoretical framework that affects enterprises' involuntary default by combining the network embeddedness theory. Previous studies mainly focused on the intentional default for the benefit of enterprises and how to manage opportunistic behavior [140,141]. Although enterprises have sufficient reasons to mitigate intentional default in the B2B OCs, it will still be affected by some unforeseen factors, which may cause enterprises to default. Research on involuntary default within the context of B2B OCs is limited. Additionally, most existing research on involuntary default has concentrated on economic difficulties, information and technology shortcomings, and environmental changes [43,49], neglecting the nature of enterprises' involuntary default due to lack of resources [52]. Enterprises' social network can be regarded as a set of resources [5,142]. Enterprises' capabilities and performance are increasingly influenced by embedded social networks. Therefore, this study adds to existing literature by highlighting the importance of structural network embeddedness on involuntary default. We acknowledge Wang et al. [17] who call for further studies into the social network and their effect in B2B OCs.

Second, this paper reveals the double-edged effect of network embeddedness. Specifically, three indicators of structural network embeddedness - density, structural holes, and centrality- exert different or even opposite effects on enterprises' involuntary default. Our empirical tests and analytical model demonstrate that density and centrality negatively affect the enterprise's involuntary default, while structural holes have a positive effect. There are some inconsistencies in these results not yet further analyzed. For example, Wang and Gong [80] and Guo et al. [77] point out that structural holes can bring a variety of information, thus improving enterprise performance. However, for most enterprises in B2B OCS, there are resource constraints and limited ability in information processing and decision-making [47], so receiving too diverse information may cause confusion and affect efficient decision-making. In general, although network embeddedness may provide resource support, it may also bring about dark sides such as information overload that contribute to enterprises' involuntary default. As observed in other studies, structural holes have a negative impact on communication and trust, and a positive effect on risk [16,82,114].

Third, this paper extends the literature on role theory by proposing the potential spillover effect of role superposition. The focal enterprise plays dual roles of businesspeople and friends in the specific seller-buyer relationship, which is called role superposition. We find the results inconsistent with other studies. The effects of dual roles of businesspeople and friends have been well documented in the fields of marketing and organizational management [27,138]. Most research shows that dual roles cause individuals' cognitive limitations and failure to meet the role expectations of group members [24], which leads to role conflict and negative effects [25]. In contrast, our study highlights the potential positive effects of dual roles, demonstrating that mutual promotion between the two roles can have beneficial outcomes. This represents an important contribution to the literature, as the moderating effect of role superposition has not been empirically examined. Notably, our study indicated that enterprises' dual roles significantly moderate the influence of density and structural holes on enterprises' involuntary default.

6.3. Practical implications

Our results provide several insights into enterprises operations within B2B OCs. First, managers should pay attention to the role of structural network embeddedness in obtaining, using, and understating information and resources, so as to alleviate involuntary default. The conclusions of this study are that network density and centrality reduce enterprises' involuntary default. Therefore, enterprises should foster a network environment with social ties, establishing close connections, occupying a favorable position, and increasing density and centrality by engaging in communication with other enterprises within the B2B OCs. In addition, our results show that structural holes promote enterprises' involuntary default. To benefit from activities within the network, enterprises should proactively cultivate specific structural network characteristics, such as high density, low structural holes, or high centrality. By doing so, enterprises can get resource support to cope with changes in market demand and unexpected supply, thus reducing involuntary default. For example, in the Alibaba merchant community, enterprises can actively post current trends, focus on the forefront of the industry, and share business experience, so as to attract more feedback from enterprises, thereby expanding their social networks and increasing centrality. By responding to the needs of other enterprises and paying attention to their feedback, enterprises can improve communication frequency and cultivate network density. This, in turn, can facilitate exchanges and transactions between enterprises, such as recommending reliable sellers to buyers, providing access to resources, or promoting the interconnection between enterprises, thereby optimizing network positions and reducing structural holes. Generally speaking, managing the structural network characteristics can help enterprises to maximize opportunities for obtaining external resources [143], reduce uncertainty with network partners during social and business activities, and receive positive feedback, ultimately leading to a reduction in involuntary default [144].

Second, the viewpoint of member network value in B2B OCs is also applicable to the choice of future partners of enterprises. Akrout and Diallo [145] emphasize the importance of investing in B2B relationships to identify reliable cooperation opportunities for trade. This strategy requires enterprises pay close attention to the activities that promote structural changes related to the network or changes in relationships with network members [66,146], so as to promote more detailed and effective information sharing from specific relationships. This discovery is in line with the trend of enterprises investing in networking, relying on network prominence to obtain resources, sharing information, and improving capabilities to facilitate transactions [84]. For enterprises engaged in B2B OCs, those with low density, more structural holes, and low centrality in networks deserve special attention, as they are more likely to engage in involuntary default.

Third, our research also suggests that dual roles of enterprises within B2B OCs, encompassing both social and business functions, merit attention. Our result show that dual roles enhance the inhibition effect of density on involuntary default and weaken the promotion of structural holes on involuntary default. Given the positive effect of enterprises' dual roles [139], we recommend that enterprises in B2B OCs carefully consider who they interact with before making decisions. Because specific binary relationships can bring in-depth and additional resources which complement the broad information from the network, thus further improving the capabilities of enterprises to respond to market demand and supply and realizing the spillover effect of role superposition. In addition, we suggest that managers representing their enterprise in B2B OCs should focus on improving their ability to balance these dual roles. Managers should learn to integrate cross-role knowledge to enhance enterprise competitiveness, rather than taking a 'free ride' based on social capital [16].

7. Limitations and future directions

This study has several imitations. First, we focus on "post-reply" networks, not those based on friendship or pure transactions. The "post-reply" network is based on interactions and emphasizes problems caused by perception and weak relationship, such as information asymmetry, which is very common in trading networks and cooperative networks. However, because our network may be based on economic participation, it may also be conducive to promoting social participation. Although these data do not represent the transactions relationships in our sample, we believe that collecting supplementary data related to transactions may help in understanding the role of other transactional relationships in the enterprise network. Second, due to the weak relationship of B2B OCs, we only consider the impact of structural network embeddedness on enterprises' involuntary default. In other OCs where members show strong relationships [57], the influence of relational embeddedness on the behavior of members should also be taken into account. Third, the problem of homophily in social networks cannot be ignored [127,128]. Although our research tries to alleviate the influence of homophily in three ways, we still need to find other ways to solve this problem in future research.

Compliance with ethical standards

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Research involving human participants and/or animals

The work does not involve human participants and animals.

Informed consent

The work did not involve informed consent.

CRediT authorship contribution statement

Xin Chen: Writing – original draft, Methodology, Conceptualization. **Wen-li Li:** Writing – review & editing, Supervision, Funding acquisition.

Disclosure of potential conflicts of interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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